

AI INFRASTRUCTURE ADVISORY

An AI data center is more than just a high-density facility. AI is an exacting application, making specific demands on infrastructure, shaping the design and technology of the facility and how it is constructed, commissioned, operated and maintained.

Many organizations need to deploy AI infrastructure quickly and economically — but without sacrificing resiliency, performance, scalability and energy efficiency. Uptime Institute AI Infrastructure Advisory enables this with five pre-packaged consulting and support services focused on each stage of the process.

To introduce some of the issues involved in delivering and operating AI data center infrastructure, Uptime Institute has published five short papers. This report is **number 2** in the series.

- 1 Design Development and Review**
- 2 Technical Vendor Requirements and Evaluation**
- 3 Construction Oversight and Validation**
- 4 Level 4 and 5 Commissioning**
- 5 Operations and Management Strategy**

Technical Vendor Requirements and Evaluation

This paper describes a strategic approach to choosing the appropriate technology, developing requests for proposal or bid tender specifications, and evaluating vendors

Having determined a design for the proposed AI facility, the next stage is to follow a carefully defined process to identify and procure infrastructure and equipment. In the fast-evolving AI data center sector, poor decision-making can have major long-term consequences. Developers need a strategy for working with vendors, selecting technology solutions and developing detailed specifications for the final delivered facility.

Project requirements and plans

Facilities for AI often need to be built quickly — but not at the expense of quality or the overall business objectives. They also need to maintain a level of resilience appropriate to the workloads they will support. This paper describes how existing construction practices need to evolve to support AI infrastructure, and how Uptime Institute's Construction Oversight and Validation helps organizations navigate these challenges.

The construction challenge

The conceptual ideas embodied in the design need to be transformed into actionable, contractual and technical documents. These documents are used to direct the process of selecting suppliers and contractors, and to agree on their roles in the project.

The aim of this phase is to deliver an integrated and functional implementation of the design that meets the owner's project requirements. A well-defined process will produce a detailed design to hand over to the construction phase. This ensures the project is delivered efficiently to budget and on schedule.

The construction industry uses generic frameworks to manage projects. The definitive Royal Institute of British Architects (RIBA) Plan of Work defines seven phases. This paper broadly aligns with RIBA Plan of Work Stages 3 and 4.

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Technical Vendor Requirements and Evaluation

Uptime Institute provides vendor-neutral assistance defining technical requirements and vendor strategies for critical capacity components through interactive workshops and expert reviews. Uptime Institute provides actionable guidance for request for proposal (RFP) development, vendor evaluation, and future-proofing AI infrastructure to ensure reliability, scalability, and alignment with business objectives.

Specifying components

The data center design specifies facility systems that provide power, cooling and related services. These include generators, uninterruptible power supplies, power distribution and switchgear, racking and cabling systems, as well as a wide range of cooling systems. The components of these services are integrated into subsystems and, in turn, the overall design.

The design of a high-density data center, and its components, will resemble traditional data center power and cooling infrastructure in many respects. The specific demands of high-density computing, however, will dictate or favor certain designs and technologies — some of which will be relatively immature.

In particular, the high heat output of high-power GPU racks may necessitate direct-to-chip liquid cooling. Air cooling is still required to cool other components, typically handling 20-30% of the heat that is not removed by cold plates.

The power draw of GPU-based systems in large language model applications will be many times higher than for traditional CPU workloads. In addition, the power demand of training applications built in clusters will fluctuate more sharply and frequently than that of conventional workloads. This needs to be a consideration in terms of electrical design and product selection.

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Requests for proposal development

Uptime Institute consultants will deliver a series of workshops to share industry best practices related to infrastructure, business objectives and vendor requirements to facilitate the development of vendor RFPs for capacity components. This includes advising on direct-to-chip, immersion and hybrid cooling solutions, which help to determine a solution that meets IT and business objectives, and assessing equipment availability to handle peak and variable workloads that align with redundancy requirements.

Cooling

Data center facilities require the removal of heat from computing systems using heat exchangers, and the transfer of the heat outside the building using facility cooling systems, usually using air or chilled water.

Inside the IT racks, air cooling is still the norm for conventional data centers. However, liquid cooling can remove heat more efficiently and is used increasingly (and is sometimes necessary) in facilities with high-density AI systems.

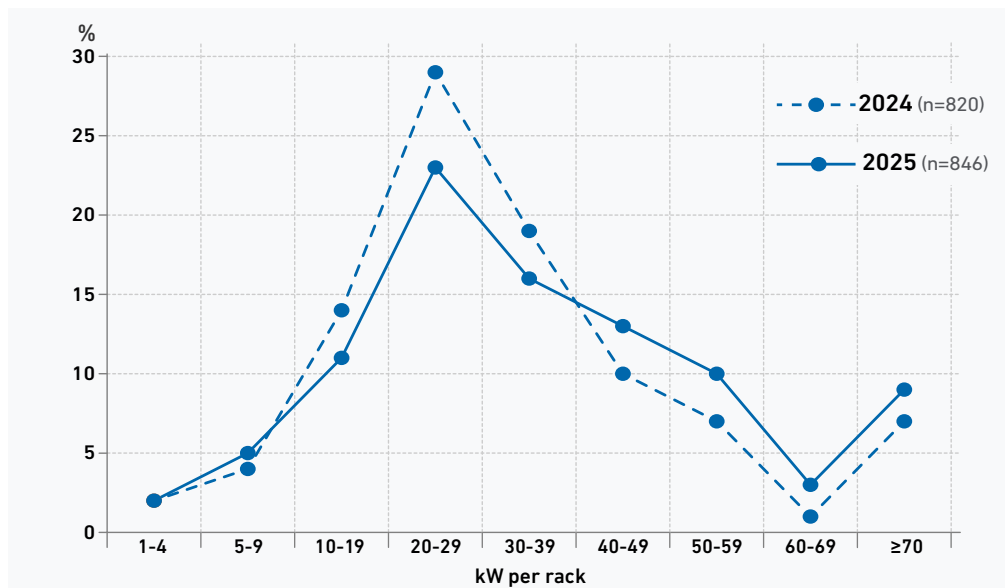


Figure 1 Liquid cooling tipping point opinions diverge and slide higher

Uptime Institute considers 40–60 kW racks to be a broad upper limit for systems that can be cooled with air alone, although the threshold for any given facility will depend on economic and business requirements, as well as current workloads, and projected future demand.

Uptime Institute’s survey responses have shown a strong preference for retaining air-cooling where possible. The perceived tipping point, above which liquid cooling is necessary, has moved higher, and early GPU-based AI systems have used air only to cool 40 kW racks.

Current generations of GPUs are increasingly dense, with the latest Nvidia racks reaching 130 kW. Uptime Institute has based its AI data center reference designs on 130 kW racks (see **Paper 1**). At this density level, there is no debate: some liquid cooling is essential, and rack level systems are shipped with integral liquid cooling.

Direct liquid cooling

Uptime Institute recommends direct-to-chip liquid (DLC) cooling for high-density data centers in most instances for several reasons. Unlike immersion cooling, which generally requires large, heavy, non-standard tanks, DLC can be deployed within the format of more-or-less standard racks, using data hall layouts resembling those used today.

This means that DLC installations can be retrofitted into existing data center spaces, subject to the physical requirements of heavier racks and any limitations on the provision of liquid cooling infrastructure (such as cooling distribution units, CDUs).

Data centers designed for DLC can include a mix of rack densities, liquid- and air-cooled racks, and are able to switch to complete air cooling if required. They can also be designed at varying levels of resiliency in accordance with Uptime Institute Tier standards.

Technology options within the DLC system include coolant selection and system design to maintain coolant quality, with leak detection and containment.

The basic design and positioning of heat exchangers/CDUs will have been determined during the design phase (see **Paper 1**). Technical specifications will be refined during the technology specification phase.

DLC does not replace air cooling. High-density racks include other components, such as networking cards, storage and auxiliary processors. These will require conventional air-cooling techniques, as well as the DLC required by components such as GPUs.

For example, a 130-kW rack will require more than 100 kW of DLC and more than 30 kW of air cooling. Uptime’s survey data has indicated that high-density facilities have specified up to 125 kW of DLC and 60 kW of air cooling per rack. Flexibility to meet even higher, or changing, demand patterns will be vital, in terms of managing future business uncertainty.

A typical design will circulate facility chilled water to both the CDUs that feed the DLC loop and the fan-walls or other air-cooling provisions.

The design must specify the topology required to provide the required level of resiliency (e.g., concurrently maintainable, fault tolerant), as well as the required specifications for the components within the whole system.

Immersion cooling

Immersion cooling is not commonly deployed today, but may be appropriate in specific instances, such as an evident long-term requirement for very high densities. Some high-performance computing (HPC) data centers have successfully used immersion cooling for many years.

However, immersion cooling represents a greater departure from current standard practices than DLC. Immersion cooling tanks are heavier than DLC racks, and staff working with immersion-cooled computing technology must follow significantly different procedures than those required for air-cooled or DLC installations.

The heavy tanks may require reinforced flooring and cannot be mixed with conventional racks as flexibly as DLC. There are, however, some technical solutions that include small immersion units installed within racks, which can be evaluated alongside other options.

Technology choices within immersion cooling include selecting a dielectric fluid that provides the required levels of performance, safety, and maintainability.

The components in the cooling infrastructure, including pumps, chillers, cooling towers, condensers, computer room air handlers, and computer room air conditioning units, need to be specified in sufficient detail for tenders.

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Cooling technology guidance

Uptime Institute consultants will provide guidance in defining cooling technology requirements for AI deployments by reviewing owner project requirements, assessing design intent, and outlining the trade-offs between direct-to-chip, immersion, and hybrid cooling solutions. Through structured workshops, technical criteria will be identified, performance expectations, and vendor evaluation considerations to support development of a robust, future-ready cooling strategy.

Power

The overall project plan requires power to be connected in phases alongside the construction — in the Uptime's reference design, this will be in phases of 15 MW (concurrently maintainable) or 12 MW (fault tolerant).

In many locations, the construction of high-density facilities is putting local grids under pressure; many are unable to provision the required power within an acceptable time, resulting in long queues. Therefore, some of the largest facilities are provisioning their own on-site primary power, for example, with gas turbines. This is a capital-intensive option with many engineering and operational challenges.

For AI data centers, resiliency assessments are needed, similarly to conventional data centers. These assessments will be determined by the needs of the business, and the level of service interruption it can tolerate. In most cases, redundant UPS, batteries, engine generators and fuel storage will still be required. Specifications should be developed to meet the power demands of high-density racks and the rapid fluctuations that are typical of GPU-based systems.

It may be necessary to distribute power at higher voltages in large AI facilities than in conventional data centers to minimize cable and/or busbar costs. This will require equipment to operate at 800 volts DC, which may be expensive and harder to source.

Similarly to cooling, power generation and distribution solutions should be selected that are compatible with likely development paths, reduce the need for early retrofits and enable reuse. For example, gas generators may be positioned as on-site primary power in one phase and later redeployed as standby power when an electrical grid connection is feasible. Another consideration is preparing power systems to support grid demand response services.

Bids and evaluation

Requests for proposal (RFP) or bid tender specifications should include all technology requirements and all the information (including financial) required to evaluate bids.

Delivery times are critically important, since long waits for equipment are common. The bid documents should set out required delivery timelines and outline the financial consequences of failing to meet these timelines. Future-proofing should be considered, including ease of upgrade and/or redeployment.

Uptime recommends evaluating three vendors for each type of equipment (including UPS, engine generators, cooling units). The process can be streamlined by pre-qualifying vendors using high-level factors, such as supported technology types and location.

The bid submission process should enable vendors to ask questions that clarify project requirements and obtain product information and details not fully captured in the design. When formal bids are received, they are usually evaluated using a template that aligns with the business requirements and how these are reflected in the design.

Once all technology and product selections are complete, they can be used to develop the contracts required for delivery and integration during construction (see **Paper 3**).

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Bidder submission evaluation

Uptime Institute consultants will deliver vendor-neutral guidance, create a draft evaluation-criteria template and participate in working sessions to provide independent bidder submission reviews, ensuring vendors align with technical requirements while maintaining full neutrality.

Partnership and support

During technology development and product selection, it is essential that the project owner maintains full control of the process.

In most cases, partners will be engaged to support parts of this process, but clear boundaries should be preserved to ensure the project owner fully understands the information required to make decisions — from basic technology direction to detailed vendor selection.

Uptime Institute AI Infrastructure Advisory: Technical Vendor Requirements and Evaluation Summary

Service overview

Uptime Institute AI Infrastructure Advisory: Technical Vendor Requirements and Evaluation helps define technical requirements and vendor strategies for critical capacity components through interactive workshops and expert reviews.

Service components

Requirements and objectives workshop for capacity components

- Development of documentation and information needed to build RFP.
- Advise on direct-to-chip, immersion, and hybrid cooling solutions; and how to determine a solution that meets the IT and business objectives.
- Availability of equipment to handle peak and variable workloads to align with redundancy requirements.
- Establish framework for vendor evaluation based on project requirements.

Bid tender specifications review

- Review capacity component equipment bid tender specifications used in the final RFP to ensure all technical requirements for projects are reflected.

Vendor evaluation

- Advise in identifying requirements for pre-selection and pre-qualification of potential bidders.
- Develop a draft evaluation criteria template.
- Identify key observations of bidder responses.

Key outcomes

- **Clear, aligned technical requirements.** Vendor solution requirements are defined from the owner project requirements to ensure cooling and power technologies meet AI workload and business objectives.
- **Robust RFP foundations.** Workshops establish the documentation, technology criteria and future-proofing considerations needed to build strong, vendor-ready technical requirements.
- **AI optimized technology strategy.** Cooling and power options are evaluated for fit, feasibility and long-term adaptability.
- **Consistent, transparent vendor evaluation.** A structured evaluation framework and template are developed to help assess bidders consistently across all capacity component types.
- **Informed vendor engagement and review.** Uptime advises on pre-qualification, participates in working sessions and bidder interviews, and provides observations on RFP responses to support well-grounded decision making.
- **Actionable recommendations for selection.** Independent findings, key observations and guidance to assist with the insights needed to confidently select vendors, while all final decisions remain entirely client-owned.



The rapid deployment of AI workloads demands infrastructure that delivers both speed to market and uncompromising resilience. Uptime Institute AI Infrastructure Advisory provide independent technical oversight across the full life cycle — from design through operations — ensuring your AI data center performs as intended from day one.

With over 30 years of experience in setting the global standard for data center performance, Uptime Institute brings unmatched expertise to the unique challenges of AI infrastructure: power densities of >40-400 kW per rack, hybrid liquid and air cooling systems and the operational complexity that mission-critical AI workloads demand. Our vendor-neutral advisors help organizations accelerate delivery while protecting investments from design drift, construction deviations and operational risks.

Whether you are building a new AI facility, retrofitting existing infrastructure or evaluating technology vendors, Uptime Institute provides the independent assurance that your infrastructure will be resilient, available and ready to support your AI ambitions. For more information visit: uptimeinstitute.com

Uptime Institute is the Global Digital Infrastructure Authority. With over 4,000 awards issued in over 122 countries around the globe, and over 1,100 currently active projects in 80+ countries, Uptime has helped tens of thousands of companies optimize critical IT assets while managing costs, resources, and efficiency. For over 30 years, the company has established industry-leading benchmarks for data center performance, resilience, sustainability, and efficiency, which provide customers assurance that their digital infrastructure can perform across a wide array of operating conditions at a level consistent with their individual business needs. Uptime's Tier Standard is the IT industry's most trusted and adopted global standard for the design, construction, and operation of data centers. Offerings include the organization's Tier Standard and Certifications, Management & Operations reviews and assessments including SCIRA-FSI financial sector risk assessment, the Sustainability Assessment, and a broad range of additional risk management, performance, availability, and related offerings. Uptime Education training programs have been successfully completed by over 100,000 data center professionals, such as the much-valued ATD (Accredited Tier Designer) and AOS (Accredited Operations Specialist). The Uptime Education curriculum has been expanded by the acquisition of CNet Training Ltd. In 2023.

Uptime Institute is headquartered in New York, NY, with offices in London, Sao Paulo, Dubai, Riyadh, and Singapore, and full-time Uptime professionals based in over thirty-four countries around the world. For more information, visit uptimeinstitute.com